

Source one-defect Gram

Euler family, Digamma–Loewner, Uvarov completion, and one null mode:
class 3 is not reached

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Abstract

Round 403 found that on FRAME-A the compression $M = P_-^{\text{ref}} B P_-^{\text{ref}}$ of the source perturbation $B = R - R^{\text{ref}}$ is positive definite (residual 0, 49/49, spectrum in $[0.023, 0.710]$), while a mask-preserving permutation of the von Mangoldt weights destroys both the Gram property and P1. The reviewer hypothesis (DCCLXIX) is that this Gram is the finite Euler-family Gram of $\Lambda(p^r) = \log p$ at $\log(p^r) = r \log p$, completed by a Digamma–Loewner block, an Uvarov off-diagonal, and exactly one null mode — a source-explicit identity $Q^T A_0 Q = FF^T - \ell \ell^T$ from which $\text{ind}_- A_0 \leq 1$ would follow by congruence, with no asymptotic analysis. Three classes are distinguished. Only class 3 (closed-form F from Λ , log, Fold, Tent, Digamma, and the explicit Bernstein–Szegő modes, breaking exactly under Λ -permutation and position jitter) would count. After one honest round, class 3 is not reached. The only exact factor of M is Cholesky of M itself (class 1). No RH claim.

Status. Lemmas 1–5 are proved (finite identities). Proposition 1 is a finite machine census that the class-3 identity fails on FRAME-A and on the small window $k_z = 3$. The stop rule of DCCLXIX fires: one round, then stop. No RH claim. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim.

1 The identity and the three classes

Write $R = R_{N-3}$ for the dual Christoffel–Darboux Gram of the first $N_w - 3$ dual OP columns on the hole nodes Y , $A_0 = R - \frac{1}{2}I$, R^{ref} for the Bernstein–Szegő Fejér dual on the same Y , and $B = R - R^{\text{ref}}$. Let Q be an orthonormal basis of the negative space of $A_{\text{ref}} = R^{\text{ref}} - \frac{1}{2}I$, and $M = Q^T B Q$. Satz G asks for a source-explicit factor

$$Q^T A_0 Q = FF^T - \ell \ell^T \quad \text{i.e.} \quad Q^T A_0 Q + \ell \ell^T = FF^T, \quad (1)$$

with F a closed formula in $(\Lambda(p^r), \log(p^r), \text{Fold}, \text{Tent}, \psi, Q)$ *before* evaluation, and ℓ the one null mode. Congruence would then give $\text{ind}_- A_0 \leq 1$ if (1) held on a full flag, not merely on a subspace.

Three classes (binding):

1. **CHOLESKY_TAUTOLOGY.** F is a Cholesky factor, a square root, or an eigenfactor of M or of A_0 , or a post-hoc solve of $XX^T = M$. Always exact when $M \succ 0$; silent under every mutant of the source that leaves M positive definite.
2. **REFERENCE_COORDINATE_ONLY.** Q is the numerical projector of A_{ref} , or a full Chebyshev- U basis of Y that contains every subspace tautologically. The factor of M remains class 1.

3. **SOURCE_GRAM_EXACT.** F is assembled from the Euler features $f_{p,r}^{c,s}(j) = \sqrt{2\log p} p^{-r/4} \cos / \sin(r\theta)$ the Cauchy features $(m + z_j)^{-1}$, Fold/Tent as an outer linear transfer, and an Uvarov stacking $[F_-; F_+]$, minus one explicit null mode. Residual 0 on MAIN; residual explodes under Λ -permutation and position jitter.

Only class 3 counts. Cholesky of M is logged as a named control, never as a construction of F .

2 Finite identities

Lemma 1 (Chebyshev addition). *Over $\mathbb{Q}$, $T_2(x)T_1(x) = (T_3(x) + T_1(x))/2$. At $x = 3/7$: $T_2 = -31/49$.*

Lemma 2 (Cauchy kernel). *Over $\mathbb{Q}$,*

$$\frac{1}{(m+z)(m+w)} = \frac{1}{z-w} \left(\frac{1}{m+w} - \frac{1}{m+z} \right) \quad (z \neq w).$$

Lemma 3 (Digamma–Loewner). *For $z, w \in \mathbb{C} \setminus \{0, -1, -2, \dots\}$,*

$$\frac{\psi(z) - \overline{\psi(w)}}{z - \overline{w}} = \sum_{m=0}^{\infty} \frac{1}{(m+z)(m+\overline{w})}.$$

At $z = \frac{1}{4} + \frac{7}{10}i$, $w = \frac{1}{4} - \frac{3}{10}i$, $\text{mpmath dps} = 40$ matches the series to $4.6 \cdot 10^{-41}$.

The identity is classical ($\psi(z) = -\gamma + \sum_m [(m+1)^{-1} - (m+z)^{-1}]$) and is *not* the claim that the archimedean block of A_0 equals this Gram.

Lemma 4 (fold is linear in lags). *The folded signed weights are the cosine transform of the lag sequence c . On FRAME-A, $\|w - (w_P + w_A)\|_2 / \|w\|_2 = 3.5 \cdot 10^{-16}$.*

Prime and archimedean contributions therefore add *before* the dual Christoffel–Darboux map. The CD Gram itself is not linear in the weights (dual weights $u^\vee \propto (1-x)^2 / (S\kappa^2)$ invert u).

Lemma 5 (the ref-negative compression is already PD). *On FRAME-A, $Q^T A_0 Q$ has $\text{ind}_- = 0$ and $\lambda_{\min} = 5.227 \cdot 10^{-3}$.*

The remaining negative of A_0 ($\text{ind}_- A_0 = 1$, $\lambda_{\min}(A_0) = -3.85 \cdot 10^{-2}$) does *not* live in the reference-negative space. Any identity of the form (1) on this Q would have to have $\ell = 0$ and $Q^T A_0 Q = FF^T$ with no defect. The one-defect, if it exists, sits in the Uvarov complement (cross terms B_{-+} and the half-modes of R^{ref}).

3 The class audit

Proposition 1 (class 3 fails). *On FRAME-A the source-explicit candidates of legs 0, 2, 3, 4, 5 do not satisfy (1). The only exact factor of M is Cholesky of M (class 1).*

Leg 0. Chebyshev- U (equivalently $\sin((k+1)\varphi)/\sin\varphi$ restricted to Y spans all of $\mathbb{R}^{104}$, hence spans the 49-space tautologically. No consecutive k -block of size 49 recovers V_n (max principal angle 90° for the first 49 and the last 49 modes). Greedy energy selection needs all 104 modes. The negative space of the explicit BS dual is not a Fourier subset of Y . The numerical projector of A_{ref} remains the control — class 2 coordinates, not an analytic Y_k .

Leg 2 (Euler). Features $f_{p,r}^{c,s}(j) = \sqrt{2 \log p} n^{-1/4} \cos / \sin((\log n / \Delta) \varphi_j)$ give a Gram whose compression to V_n has residual 209 against M , with Frobenius alignment 0.767 after the fact. A fitted global scale still leaves residual 0.642. The addition theorem is true as trigonometry (Lemma 1) and the tent/cosine transform is true as the definition of the lag map (Lemma 4); neither survives the dual CD Gram. Fold/Tent is an outer *linear* transfer at the weight layer, not at A_0 .

Leg 3 (Digamma). Lemma 3 holds on the function. The Cauchy–Loewner Gram of $z_j = \frac{1}{4} + \frac{1}{2}i \varphi_j$ has residual 207 against A_0 and alignment 0.098 against M . The archimedean block of A_0 is not this Loewner matrix. A Digamma shift $\frac{1}{4} \rightarrow \frac{3}{4}$ stays at residual 38 (still $O(1)$).

Leg 4 (Uvarov). Omitting the Gram part of B on the negative space returns $\text{ind}_- = 49$; keeping only that Gram leaves $\text{ind}_- = 13$ (r403). The cross terms carry $\|B_{\text{cross}}\|_F / \|B\|_F = 0.617$ and numerical rank 104. Stacked tent or Euler features on (Y, Y^c) do not reproduce B (YY-block residual 1 and 266). “Only-Gram leaves 13” is the incomplete-PSD phenomenon of r403, not the off-diagonal of a source factor.

Leg 5 (null mode). $|\cos|(v_-, 1) = 0.007$; $|\cos|(v_-, V_n) = 0.762$ and $|\cos|(v_-, V_n^\perp) = 0.648$ — the defect is mixed. The candidate $\ell = \sqrt{\sum_{p \leq e^{2\alpha}} \log p} Q^T 1$ has cosine 0.069 with the lowest mode of $Q^T A_0 Q$ (which is already positive). The ones-mode is not the null mode.

4 Mutant table

A class-3 identity must break loudly under source mutants. Because no class-3 identity is present, residuals stay $O(1)$. Cholesky of M stays exact on MAIN and ceases to exist as soon as M leaves the PD cone.

mutant	Euler/Cauchy residual vs M	Cholesky of M
MAIN (von Mangoldt)	209 (align 0.767)	$2.3 \cdot 10^{-16}$
Λ -permutation	314	undefined (M has 6 negatives)
position jitter 10^{-3}	209	—
$\Lambda(p^r) \mapsto \log(p^r)$	249	—
Digamma $\frac{1}{4} \rightarrow \frac{3}{4}$	38	—
omit null / omit Uvarov	still 209	—

The Λ -permutation ratio $314/209 \approx 1.5$ is not an explosion from residual 0. It is a reshuffling of an already failed candidate. Cholesky, by contrast, is bit-exact on every world in which M happens to be PD — including none of the mask-preserving permutations of FRAME-A.

5 What is missing

The dual CD Gram is a nonlinear function of the folded weights. Legs 2–3 factor the *linear* lag/weight layer (Lemmas 1–4), which is the right algebra for the spectral density and the wrong algebra for A_0 . An exact source factor of A_0 would have to intertwine the Borodin dual inversion $u \mapsto u^\vee$ and the depth- $(N_w - 3)$ OP recurrence with the Euler/Digamma Gram — a dictionary that is not in the r342 Digamma/tent list and is not produced by restricting Chebyshev- U to Y . The compression $Q^T A_0 Q$ being already PD (Lemma 5) further shows that the one defect of MAIN does not sit where the reviewer placed the null mode. The stop rule

forbids iterating a Cholesky factor, a numerical square root of M , or a fitted scale of the Euler Gram (alignment 0.767 is a census, not a construction).

Machine check

Standalone: Chebyshev addition, Cauchy kernel, Loewner identity, lag-linearity of the fold. Construction: FRAME-A inertia and PD compression, Fourier subset, Euler/Loewner residuals, only-Gram and cross share, Cholesky tautology versus permute, ones-mode, Λ -perm mutant. Discovery census: `one_defect_gram_probe.py` (`--smoke` 19/19 / full 21/21), SPEC_SHA c0260053d759bd14, wall 0.2 s. Exit line: ONE DEFECT GRAM VERIFIED.

Claim boundary

Finite identities for addition, the Loewner series, lag-linearity, and a PD compression; a named census that the source-explicit one-defect factor is not class 3 on the tested windows. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, G_ε^μ / *Bernstein–Szegő Fejér*, sealed probe, August 2026.
- [2] S. Hamann, *P1 is not a construction theorem*, standalone note, August 2026, `rh/problem/p1_construction.tex`.
- [3] S. Hamann, *Bulk one-defect*, standalone note, August 2026, `rh/problem/bulk_one_defect.tex`.